

Properties of Propositional Logic and Predicate Calculus

a. Equivalences as properties in Propositional Logic

1. $(p \Rightarrow q) \equiv ((\sim p) \vee q)$
2. $(p \Rightarrow q) \equiv (\sim q \Rightarrow \sim p)$
3. $(p \Leftrightarrow q) \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$
4. $\sim (p \Leftrightarrow q) \equiv (p \wedge \sim q) \vee (q \wedge \sim p)$
5. $\sim (p \Leftrightarrow q) \equiv (p \wedge \sim q) \vee (q \wedge \sim p)$

b. Equivalences as properties in Predicate Calculus

1. $\sim (\forall x P(x)) \equiv \exists x (\sim P(x))$
2. $\sim (\exists x P(x)) \equiv \forall x (\sim P(x))$
3. $\exists x (\sim P(x) \Rightarrow Q(x)) \equiv \forall x P(x) \Rightarrow \exists x Q(x)$
4. $\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$
5. $\forall x (P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)$
6. $((\forall x P(x)) \vee (\forall x Q(x))) \Rightarrow \forall x (P(x) \vee Q(x)) : \text{Tautology}$
7. $\exists x (P(x) \wedge Q(x)) \Rightarrow \exists x P(x) \wedge \exists x Q(x) : \text{Tautology}$

Consider statements corresponding the propositions **p** and **q**. Also consider a set, say **A**, and predicates **P** and **Q** on **A**.

Write the sentences/statements corresponding all the above well-formed formulae **a 1.- 5.** and **b 1. – 7.**